



Final Year Individual Research Project

Guidelines

Module Title: Final Year Individual Project

Module Code: COM 4901

Applicable to:

Bachelor of Science (Honours) Degree Programmes

- Software Engineering**
- Computer Networks and Cyber Security**
- Management Information Systems**
- Data Science**

Faculty of Computer Science and Engineering

KIU University

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1. Introduction

The Final Year Individual Research Project (COM4901) is a compulsory capstone module offered by the Faculty of Computer Science and Engineering at KIU. This project provides students with the opportunity to demonstrate independent research capability, professional responsibility, and technical competence by planning, developing, implementing, and evaluating an original project in an appropriate computing domain.

Students are expected to apply knowledge and skills acquired throughout the programme and demonstrate the ability to engage in systematic research and development work at an undergraduate level. This guideline document serves as a formal reference for students, supervisors, and the faculty to ensure consistent academic quality, fairness, and transparency in project execution and assessment.

2. Learning Outcomes of the Module

At the end of this module, the student should be able to:

LO1: develop a project proposal to solve a problem in a relevant area/domain using knowledge and skills acquired in the academic program.

LO2: identify ethical or professional issues relevant to the project topic.

LO3: carry out literature survey related to the topic.

LO4: develop a methodology to design and implement a solution.

LO5: implement, test, evaluate and validate the solution / prototype.

LO6: demonstrate oral and written communication skills in the preparation of prototypes, demos, and documentation that meet professional standards.

3. Types of Projects

Students may undertake either a Research-Oriented Project or a Development-Oriented Project based on the nature of the selected topic. Regardless of the project type, all projects must demonstrate appropriate planning, academic integrity, professional conduct, and clear evaluation/validation of outcomes.

3.1 Research-Oriented Project

A research-oriented project typically involves investigating a research problem or gap through a structured methodology. It normally includes a research question/objectives, literature-based justification, and experimental evaluation (e.g., benchmarking, comparison studies, statistical analysis) leading to research findings and conclusions.

Examples: machine learning model comparison, cyber security evaluation frameworks, predictive modelling studies, algorithm design and benchmarking.

3.2 Development-Oriented Project

A development-oriented project typically involves designing and implementing a computing solution to address a real-world problem. It normally includes requirements analysis, system design/architecture, implementation, testing, and evaluation/validation of the proposed solution.

Examples: web/mobile applications, IoT systems, dashboards, platforms, software prototypes, automation solutions.

The examples provided are indicative only and do not limit the range of topics permitted under COM4901.

4. Roles and Responsibilities

4.1 Student Responsibilities

Each student is expected to:

- Select an appropriate project topic aligned with the computing discipline and module learning outcomes.
- Obtain an approved supervisor (internal or external) in accordance with Faculty policy.
- Submit the project proposal and complete all required presentations and reviews on time.
- Maintain continuous progress through regular meetings with the supervisor (minimum frequency as defined by the Faculty).
- Maintain a Project Diary/Logbook including meeting records, progress updates, action plans, and supporting evidence.
- Follow ethical practices and demonstrate professional responsibility throughout the project lifecycle.
- Ensure academic integrity in all submissions, including correct citation and avoidance of plagiarism.
- Prepare and submit interim reports, final dissertation, and all supporting materials by the given deadlines.
- Prepare and deliver the final presentation and participate in viva/defence professionally.
- Ensure proper backup and version control of project files (e.g., repository usage, cloud storage).

4.2 Supervisor Responsibilities

The supervisor plays the primary mentoring role and is expected to:

- Guide the student in refining the topic, defining the problem statement, and formulating measurable objectives.
- Provide academic guidance on research direction, literature review, methodology, and technical design.

- Monitor student progress continuously and provide timely feedback to improve quality and feasibility.
- Conduct regular supervision meetings and review the student's diary/logbook and progress evidence.
- Review the quality of documentation (proposal, interim report, dissertation) and provide improvement comments.
- Ensure that the student adheres to academic integrity, ethical standards, and professional conduct.
- Identify risks early (delays, scope issues, technical barriers) and advise corrective actions.
- Report serious progress issues or misconduct concerns to the Project/Module Coordinator where necessary.

An internal or external supervisor must have relevant academic qualifications and/or industry experience in the project domain. External supervisors must hold at least an undergraduate degree in Computing (or a closely related discipline) and should have a minimum of two years of relevant professional experience. All external supervisors must be formally approved by the Project/Module Coordinator prior to commencement of supervision.

4.3 Faculty Coordinator Responsibilities

The Project Coordinator / Module Coordinator is responsible for ensuring that the COM4901 project module is implemented consistently and fairly across all students in accordance with Faculty and University policies.

The coordinator's responsibilities include:

- **Module governance and communication:** communicate project timelines, submission deadlines, required templates, and assessment requirements to students and supervisors.
- **Official topic confirmation:** ensure that project topics are first discussed and approved by the assigned supervisor (academic approval), and then provide final confirmation/registration of topics to ensure alignment with module learning outcomes, faculty standards, and policy requirements.
- **Supervisor validation and allocation:** approve internal/external supervisors based on eligibility requirements and ensure fair workload distribution.
- **Progress monitoring:** monitor student progress at module level and intervene when delays, risks, or non-compliance are identified (e.g., missed meetings, lack of progress evidence).
- **Coordination of assessments:** schedule and coordinate proposal presentations, interim reviews, final presentations, and viva examinations.
- **Examiner / panel appointment:** appoint assessment panels/examiners, ensure moderation where required, and support consistent assessment practices across groups.
- **Administrative decisions:** manage requests such as topic changes, supervisor changes, and extension requests in accordance with Faculty procedures.

- **Academic integrity and ethics oversight:** ensure plagiarism checking and ethical compliance processes are followed and report academic misconduct if detected.
- **Documentation and record management:** maintain official records of approvals, assessment decisions, marking reports, and final project outcomes.

5. Project Process and Timeline

Final Year Individual Project will be conducted across the academic year following structured milestones.

The standard milestones include:

1. Topic Selection & Supervisor Confirmation
2. Proposal Submission
3. Proposal Presentation
4. Project Execution & Monitoring (with diary/logbook updates)
5. Interim Progress Review
6. Final Thesis Submission
7. Final Presentation
8. Viva / Dissertation Defence
9. Corrections & Final Submission (if required)

5.1 Assessment Components and Weightage

The assessment of COM4901 is based on continuous evaluation throughout the project lifecycle. The assessment components and corresponding weightage are summarized below.

Assessment Component	Weightage (%)
Proposal Writing	15
Presenting the Proposal	10
Interim Report	15
Submission of Final Report	30
Presentation and Demonstration of the Project	30
Total	100

All assessment components must be completed to successfully pass the module.

6. Proposal Guidelines

6.1 Purpose of the Proposal

The proposal is a formal document outlining the project's topic, objectives, justification, methodology, and implementation plan. It serves as the foundation for approval and evaluation.

6.2 Proposal Format Requirements

The proposal must follow academic formatting standards:

- **Font:** Times New Roman
- **Font Size:** 12
- **Line Spacing:** 1.5
- **Margins:** 1 inch (all sides)
- **Page Limit:** Maximum 10 pages (excluding references and appendices)

6.3 Proposal Structure

A proposal should contain:

1. Title Page
2. Introduction / Background
3. Problem Statement
4. Research Aim and Objectives
5. Scope and Deliverables
6. Literature Summary (brief overview)
7. Proposed Methodology / System Approach
8. Tools & Resources Required
9. Project Plan / Timeline (Gantt Chart recommended)
10. References (IEEE format)

6.4 Proposal Presentation

Students must present the proposal to a panel. The presentation should include:

- Project motivation and background,
- Clear problem definition,
- Objectives and expected deliverables,
- Methodology or system design approach,
- Feasibility and project plan.

7. Progress Monitoring & Diary Requirement

Continuous monitoring is mandatory. Every student must maintain a **Project Diary / Logbook**, which serves as evidence of progress.

The diary must include:

- meeting dates and attendance,
- supervisor feedback and action points,
- completed tasks and outcomes,
- challenges faced and how they were handled,
- screenshots, research notes, diagrams (where relevant),
- weekly progress summaries.

Students must meet supervisors regularly (minimum once every two weeks) and obtain supervisor verification/signature in the diary.

8. Ethical Considerations and Approval Requirements

Ethical and professional considerations are a mandatory requirement in COM4901, as projects may involve data collection, human participation, personal information, sensitive systems, or security-related experimentation. Students must identify ethical or professional issues relevant to the selected topic and ensure compliance with KIU policies and applicable legal requirements.

Projects that involve any of the following require ethical approval and supervisor confirmation prior to data collection or implementation:

- Human participants (questionnaires, surveys, interviews, focus groups)
- Collection or processing of personal data (names, contact details, identification numbers, etc.)
- Biometric or physiological data (e.g., EEG, facial images, voice recordings)
- Camera-based data collection or surveillance-related data
- Medical/health-related data
- Any vulnerable groups (children, patients, etc.)

Where applicable, students must obtain informed consent from participants and ensure confidentiality, anonymisation, and secure storage of collected data. All datasets and third-party resources must be used legally with proper citation and licensing compliance.

For cyber security-related projects, students must ensure that all testing and experimentation are performed only in controlled environments (e.g., lab simulations, virtual machines, authorised test

systems). Activities involving unauthorised access, hacking, exploitation of real systems, or illegal data acquisition are strictly prohibited.

9. Interim Progress Review and Interim Report

An Interim Progress Review is conducted to evaluate the progress of the project and provide structured feedback before final submission. As per the module specification, students are required to submit an Interim Report, which contributes 15% to the final module grade.

The Interim Review and Interim Report are used to assess the student's progress against planned objectives, identify risks or delays, and ensure the project remains feasible and academically sound.

9.1 Interim Report Submission Requirement

Each student must submit an Interim Report demonstrating project progress up to the interim stage. The report must provide evidence of work completed, literature progress, methodology/design, implementation status, and an updated plan for completion.

9.2 Interim Report Format and Structure

The Interim Report should be prepared following the same formatting guidelines as the final dissertation (Times New Roman 12, 1.5 spacing, 1-inch margins, IEEE referencing where applicable).

Recommended length: **10–20 pages** (excluding references and appendices).

Interim Report Structure

The Interim Report should contain the following sections:

1. Title Page

- Student name, ID
- Project title
- Supervisor name
- Module code: COM4901
- Faculty of Computer Science and Engineering, KIU
- Submission date

2. Introduction

- Background and context
- Problem statement
- Project aim and objectives

3. Progress Summary

- Tasks completed so far
- Current project status (what has been achieved)
- Evidence of progress (brief description of artefacts developed)

4. Literature Review Progress

- Summary of key literature identified
- Theoretical/conceptual foundation
- Research gap justification (if applicable)

5. Methodology / Solution Approach

- Research methodology OR system development methodology
- Data collection approach / system requirements approach as applicable
- Tools, techniques, technologies selected and justification

6. Design and Implementation Progress

- Architecture / design diagrams (if applicable)
- Implementation progress (modules completed, functions developed, prototypes)
- Screenshots/outputs where relevant

7. Testing / Evaluation Plan (Draft)

- Proposed testing strategy
- Evaluation metrics or validation plan
- Planned experiments or performance measures

8. Challenges and Risk Management

- Issues faced (technical, time, resources, data limitations)
- Actions taken to resolve challenges

- Risk mitigation plan

9. Revised Work Plan

- Updated Gantt chart / timeline
- Remaining tasks
- Milestones up to final submission

10. Conclusion

- Summary of progress
- Confirmation of feasibility and completion plan

11. References

- IEEE format

12. Appendices (Optional)

- Screenshots
- Sample data
- Meeting minutes
- Diary extracts

9.3 Interim Report Submission Requirements

The Interim Report submission must include:

- Interim Report (PDF)
- Interim Presentation Slides (if required)
- Evidence of progress (e.g., screenshots, GitHub repository link, prototype demo link)
- Project diary/logbook (if requested by panel)

9.4 Interim Review Assessment Areas

The Interim Review will typically evaluate:

- Completion level against objectives
- Progress in literature survey and research understanding
- Methodology suitability and correctness
- Implementation/design progress and evidence

- Quality of documentation and academic writing
- Updated plan and feasibility for final delivery

10. Dissertation / Thesis Guidelines

10.1 General Thesis Requirement

Each student must submit a well-structured dissertation demonstrating:

- academic writing quality,
- Research integrity,
- correct referencing,
- complete project lifecycle documentation.

10.2 Formatting Requirements

All final dissertations must follow:

- **Font:** Times New Roman
- **Size:** 12
- **Spacing:** 1.5
- **Margins:** 1 inch
- **Referencing Style:** IEEE referencing

10.3 Standard Dissertation Structure

A typical dissertation must include:

1. Title Page
2. Abstract
3. Acknowledgement
4. Table of Contents
5. List of Figures
6. List of Tables
7. Chapter 1 – Introduction
8. Chapter 2 – Literature Review
9. Chapter 3 – Methodology / Design
10. Chapter 4 – Implementation
11. Chapter 5 – Testing & Evaluation
12. Chapter 6 – Conclusions and Future Work
13. References
14. Appendices (if applicable)

11.Academic Integrity & Plagiarism Policy

All submitted documents must follow academic integrity standards. Plagiarism is strictly prohibited.

- Dissertation and proposal must be submitted through plagiarism checking tools (e.g., Turnitin).
- If similarity is unacceptably high, the student must revise and resubmit.
- Any confirmed academic misconduct will be handled according to KIU disciplinary policy.

12.Presentation and Viva (Dissertation Defence)

The Final Presentation and Viva assess:

- student's technical understanding,
- justification of project design decisions,
- demonstration of implementation and results,
- ability to answer examiner questions,
- academic maturity and professional communication.

Students must bring:

- presentation slides,
- system demo (if applicable),
- evaluation results,
- dissertation document,
- diary evidence if requested.

13.Final Submission Requirements

Final submission must include:

- Final dissertation (PDF and Word format)
- Presentation slides (PPT/PDF)
- Source code / executable / deployment file
- Datasets or experiments logs (where applicable)
- Plagiarism report
- Project diary/logbook

14.Recommendations for Research Publication and Professional Outputs

Students are strongly encouraged to produce research and professional outputs beyond the dissertation in order to enhance academic quality, industry relevance, and future employability. Where possible, students should aim to present their work in a form suitable for publication and dissemination, subject to supervisor guidance.

Where the project demonstrates strong novelty, evaluation quality, and academic contribution, students may be encouraged to submit the work to a suitable conference, symposium, journal, or institutional research forum, in accordance with Faculty procedures.

15.Conclusion

The COM4901 Final Year Project is a major academic undertaking. Students are expected to demonstrate independent research skills, professional conduct, and technical competency. The Faculty of Computer Science and Engineering ensures academic quality through structured supervision, evaluation milestones, and assessment rubrics. All students must strictly adhere to the guidelines and deadlines stated in this document.